VSDBabySoC

Physical Design and Routed Static Timing Analysis

Floorplan • Placement • Routing • Independent Sky130 HD Corners

Design flow: Synthesis → Floorplanning → Placement → Routing → Post-route STA

This report documents the physical-design screenshots and the independent timing-corner results supplied for the VSDBabySoC design.

Timing inputs recorded

• Routed database: `results/sky130hd/vsdbabysoc/base/6_final.odb` (~145 MB)

• Extracted parasitics: `results/sky130hd/vsdbabysoc/base/6_final.spef` (~17 MB)

• Timing constraints: `results/sky130hd/vsdbabysoc/base/5_route.sdc` (~839 bytes)

• Liberty libraries: 16 individual Sky130 HD PVT files, each analyzed in a separate run

• Final route SDC shown in the session defines an 11.000 ns clock and propagated clock timing.

1. Floorplanning

Floorplanning establishes the core area and initial physical organization of the design. It includes placing macros, defining rows and tracks, and preparing the layout for power distribution and standard-cell placement. The supplied screenshot is OpenROAD stage **2_floorplan**.

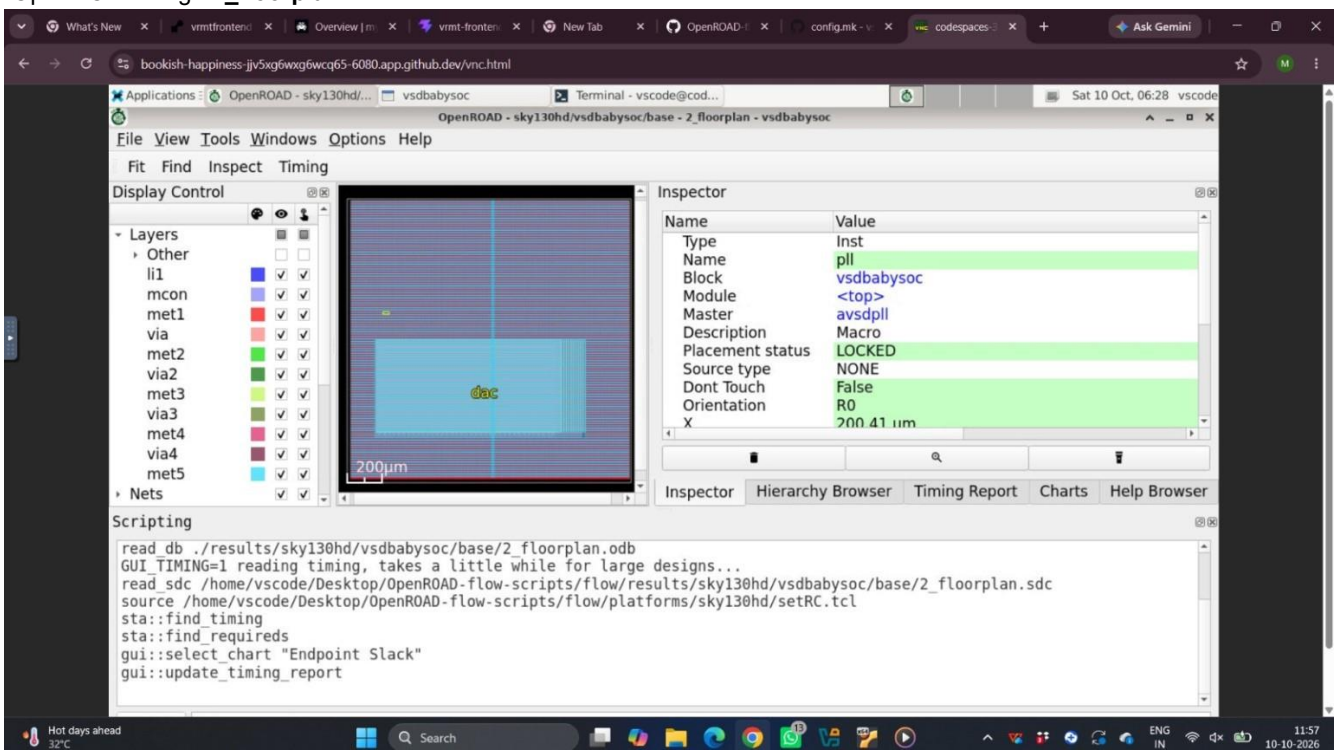


Figure 1. VSDBabySoC floorplan in OpenROAD. The PLL and DAC appear as macro instances; the inspector identifies the selected instance as `pll`, with master `avsdpll` and placement status LOCKED.

What is visible

- The macro labeled **dac** is visible in the floorplan view, with the PLL macro selected in the inspector.
- The layer panel lists routing layers and vias (li1, met1–met5, and via layers).
- The floorplan-stage console reads `2_floorplan.odb` and `2_floorplan.sdc`, sources the Sky130 RC setup, then runs timing update commands.
- The PLL placement is shown as LOCKED, which is consistent with a fixed macro placement at this stage.

Floorplanning is not detailed routing: the view primarily communicates macro/core organization and physical design setup.

2. Placement

Placement assigns standard cells to legal sites inside the floorplan. The placer aims to satisfy placement constraints while considering timing, congestion, and wirelength. The supplied screenshot is OpenROAD stage **3_place**.

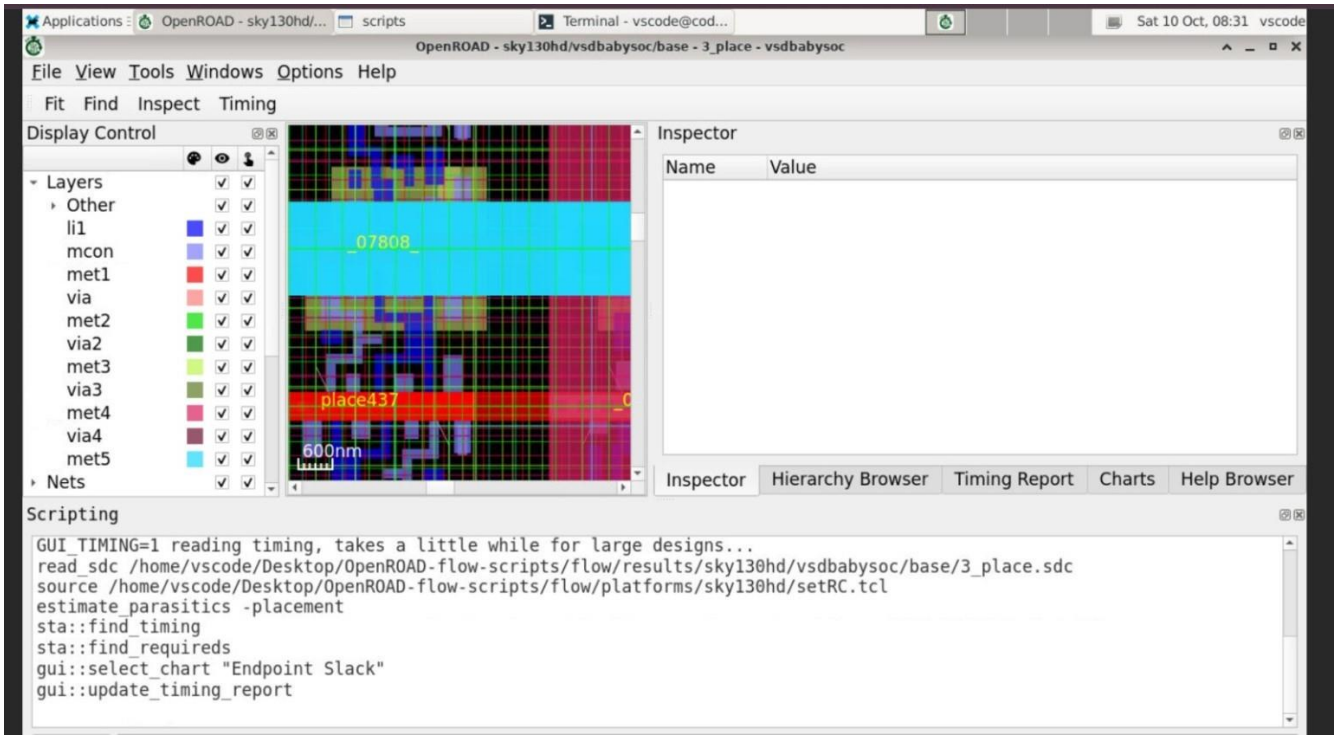


Figure 2. VSDBabySoC placement-stage OpenROAD view. The selected standard-cell instance is labeled `plade437`; the console reads `3_place.sdc` and runs placement-stage parasitic estimation and timing updates.

What is visible

- The selected cell is shown in the inspector area, with routing layers and vias enabled in the layer panel.
- The console reads `3_place.sdc`, sources the Sky130 RC setup, runs `estimate_parasitics -placement`, and updates timing.
- Placement-stage parasitics are estimates. They are not equivalent to the extracted routed parasitics used in finalpost-route STA.

The screenshot is a local zoom into the placed design, so it is useful as evidence of the placement stage but does not show the entire chip.

3. Routing and final routed database

Routing creates the detailed metal interconnect and vias that connect placed cells and macros. The final routed database is then used with extracted parasitics for post-route timing analysis.

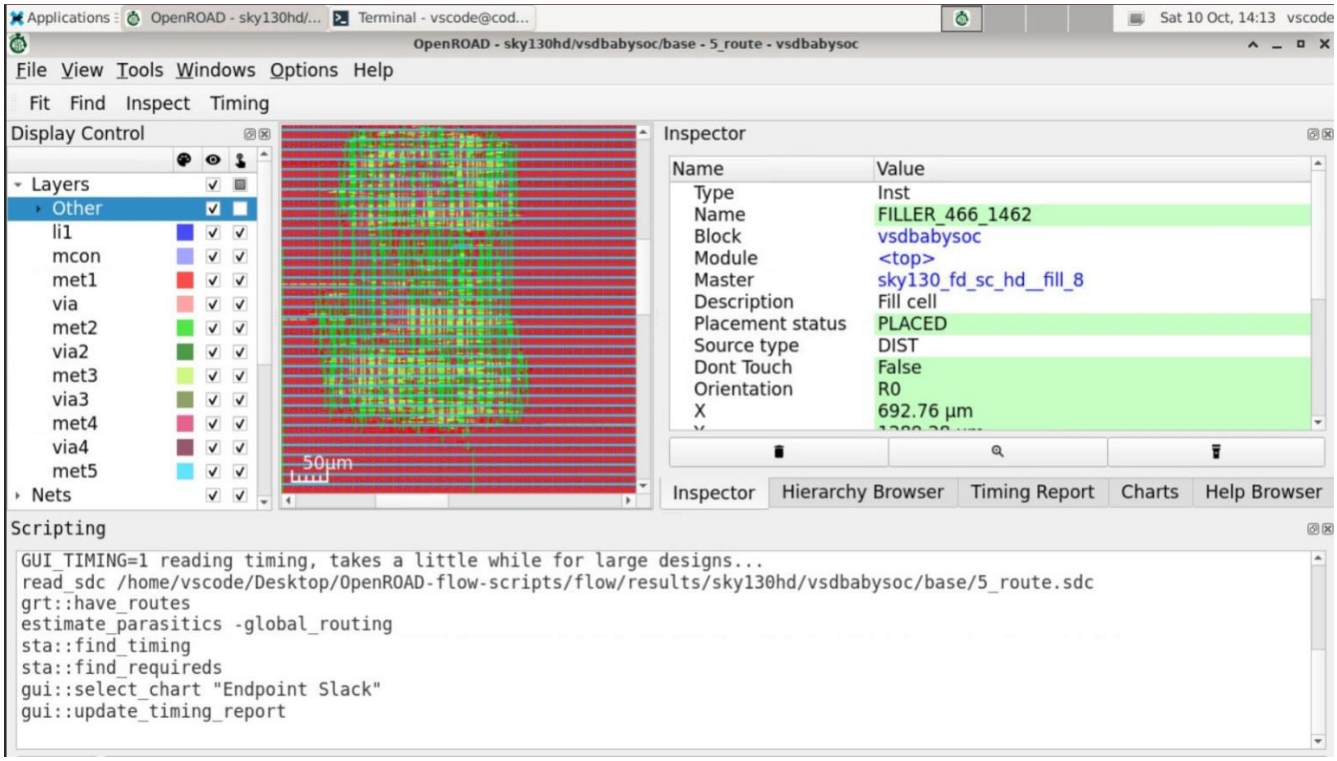


Figure 3. OpenROAD GUI showing the routed VSDBabySoC layout and visible metal/via layers. The inspector identifies a placed filler cell; the view demonstrates the routed physical database.

Files used for independent post-route STA

- `6_final.odb` — final routed OpenDB database.
- `6_final.spef` — extracted interconnect parasitics.
- `5_route.sdc` — route-stage timing constraints.
- One Liberty `.lib` file per independent run.

The timing runs are independent single-library comparisons. Their results are not a combined MCMM run and should not automatically be presented as signoff-qualified PVT coverage.

4. Setup timing results by independent corner

WNS is worst negative slack and TNS is total negative slack. Negative values indicate setup violations under the given timing model and constraints. Values below are transcribed from the results supplied in the conversation.

Library corner	Setup WNS (ns)	Setup TNS (ns)	Worst setup slack (ns)
sky130_fd_sc_hd__ff_100C_1v65	0.00	0.00	7.27
sky130_fd_sc_hd__ff_100C_1v95	0.00	0.00	8.15
sky130_fd_sc_hd__ff_n40C_1v56	0.00	0.00	5.89
sky130_fd_sc_hd__ff_n40C_1v65	0.00	0.00	6.67
sky130_fd_sc_hd__ff_n40C_1v76	0.00	0.00	7.32
sky130_fd_sc_hd__ff_n40C_1v95	0.00	0.00	8.06
sky130_fd_sc_hd__ss_100C_1v40	-3.34	-462.24	-3.34
sky130_fd_sc_hd__ss_100C_1v60	0.00	0.00	1.29
sky130_fd_sc_hd__ss_n40C_1v28	-32.03	-15425.64	-32.03
sky130_fd_sc_hd__ss_n40C_1v35	-17.90	-7390.62	-17.90
sky130_fd_sc_hd__ss_n40C_1v40	-12.08	-3942.03	-12.08
sky130_fd_sc_hd__ss_n40C_1v44	-8.72	-2114.08	-8.72
sky130_fd_sc_hd__ss_n40C_1v60	-1.00	-19.01	-1.00
sky130_fd_sc_hd__ss_n40C_1v76	0.00	0.00	2.52
sky130_fd_sc_hd__tt_025C_1v80	0.00	0.00	6.16
sky130_fd_sc_hd__tt_100C_1v80	0.00	0.00	6.27

Key observation: the most severe supplied setup result is `ss_n40C_1v28`: WNS −32.03 ns and TNS −15425.64 ns. The `ss_100C_1v40` corner also reports a violation (WNS −3.34 ns; TNS −462.24 ns).

A zero WNS/TNS summary does not mean the worst path slack is zero; it means the summary contains no negative setup slack. Keep worst-path slack in a separate column.

5. Setup path interpretation

The supplied worst setup-path report for `ss_n40C_1v28` identifies a register-to-register path in the CPU logic.

• Startpoint: `core.CPU_src1_value_a3[14]` flip-flop.

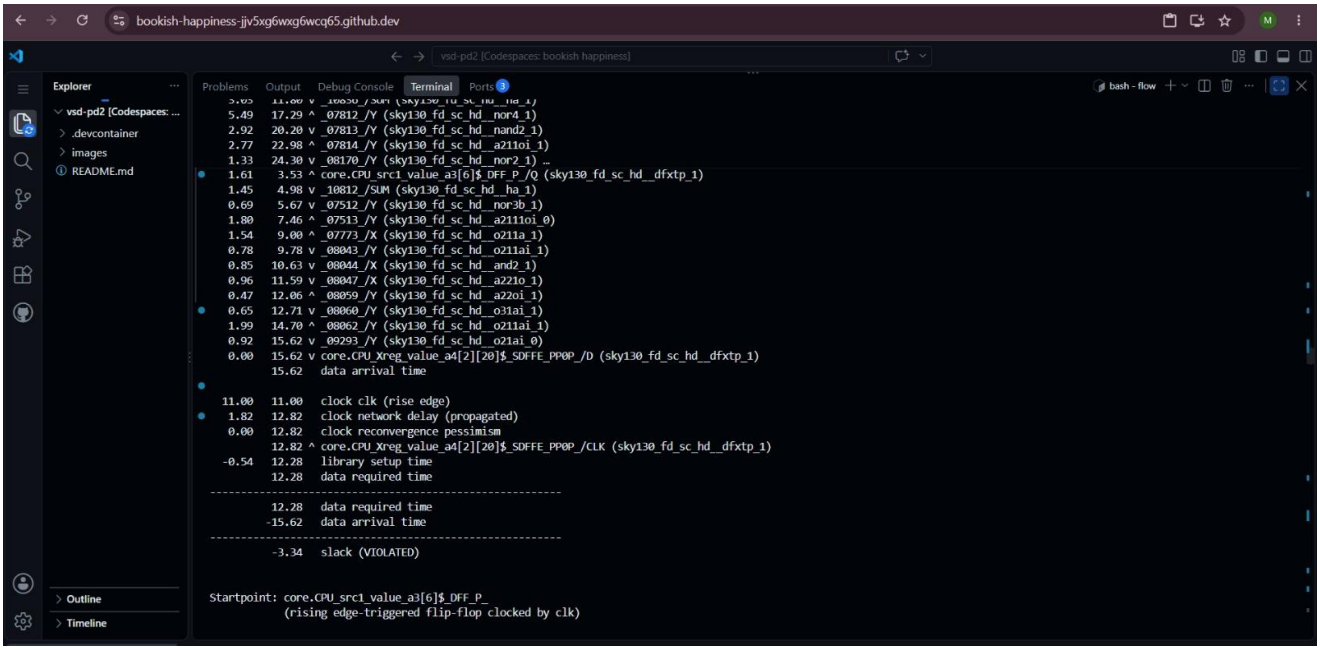
• Endpoint: `core.CPU_Xreg_value_a4[10][27]` flip-flop.

• Data arrival time: 46.45 ns; data required time: 14.42 ns.

• Slack = required time − arrival time = 14.42 − 46.45 = −32.03 ns (VIOLATED).

The path contains a sequence of combinational cells, including half-adder, NOR, NAND, AOI/OAI-style gates. The listed incremental delays accumulate into a late data arrival. This points to a long combinational path under that particular slow/low-voltage model.

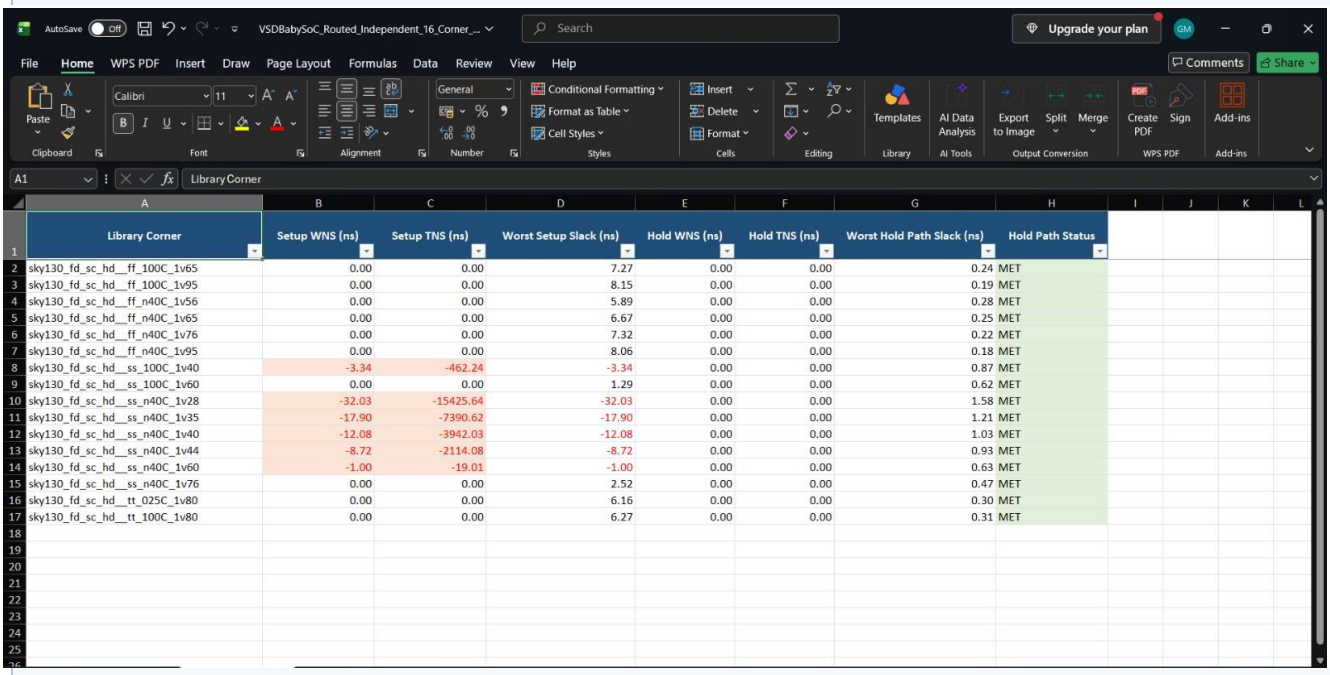
The second supplied setup screenshot (`ss_100C_1v40`) shows arrival time 15.62 ns, required time 12.28 ns, and slack −3.34 ns.



6. Hold timing results

Hold analysis checks whether data arrives too soon after the capture clock edge. The user-provided extraction from each log reports a positive worst hold-path slack and status MET for all 16 corners.

Why Hold WNS/TNS are 0.00: WNS/TNS summarize negative slack. When all hold paths pass, there may be no negative hold



Library Corner	Setup WNS (ns)	Setup TNS (ns)	Worst Setup Slack (ns)	Hold WNS (ns)	Hold TNS (ns)	Worst Hold Path Slack (ns)	Hold Path Status
sky130_fd_sc_hd__ff_100C_1v65	0.00	0.00	7.27	0.00	0.00	0.24	MET
sky130_fd_sc_hd__ff_100C_1v95	0.00	0.00	8.15	0.00	0.00	0.19	MET
sky130_fd_sc_hd__ff_n40C_1v56	0.00	0.00	5.89	0.00	0.00	0.28	MET
sky130_fd_sc_hd__ff_n40C_1v65	0.00	0.00	6.67	0.00	0.00	0.25	MET
sky130_fd_sc_hd__ff_n40C_1v76	0.00	0.00	7.32	0.00	0.00	0.22	MET
sky130_fd_sc_hd__ff_n40C_1v95	0.00	0.00	8.06	0.00	0.00	0.18	MET
sky130_fd_sc_hd__ss_100C_1v40	-3.34	-462.24	-3.34	0.00	0.00	0.87	MET
sky130_fd_sc_hd__ss_100C_1v60	0.00	0.00	1.29	0.00	0.00	0.62	MET
sky130_fd_sc_hd__ss_n40C_1v28	-32.03	-15425.64	-32.03	0.00	0.00	1.58	MET
sky130_fd_sc_hd__ss_n40C_1v35	-17.90	-7390.62	-17.90	0.00	0.00	1.21	MET
sky130_fd_sc_hd__ss_n40C_1v40	-12.08	-3942.03	-12.08	0.00	0.00	1.03	MET
sky130_fd_sc_hd__ss_n40C_1v44	-8.72	-2114.08	-8.72	0.00	0.00	0.93	MET
sky130_fd_sc_hd__ss_n40C_1v60	-1.00	-19.01	-1.00	0.00	0.00	0.63	MET
sky130_fd_sc_hd__ss_n40C_1v76	0.00	0.00	2.52	0.00	0.00	0.47	MET
sky130_fd_sc_hd__tt_025C_1v80	0.00	0.00	6.16	0.00	0.00	0.30	MET
sky130_fd_sc_hd__tt_100C_1v80	0.00	0.00	6.27	0.00	0.00	0.31	MET

slack to sum, so both values are zero even though the worst hold path has a positive slack (for example, +1.21 ns).

Example from `ss_n40C_1v35`: data arrival = 4.38 ns, required time = 3.17 ns, and slack = +1.21 ns (MET).

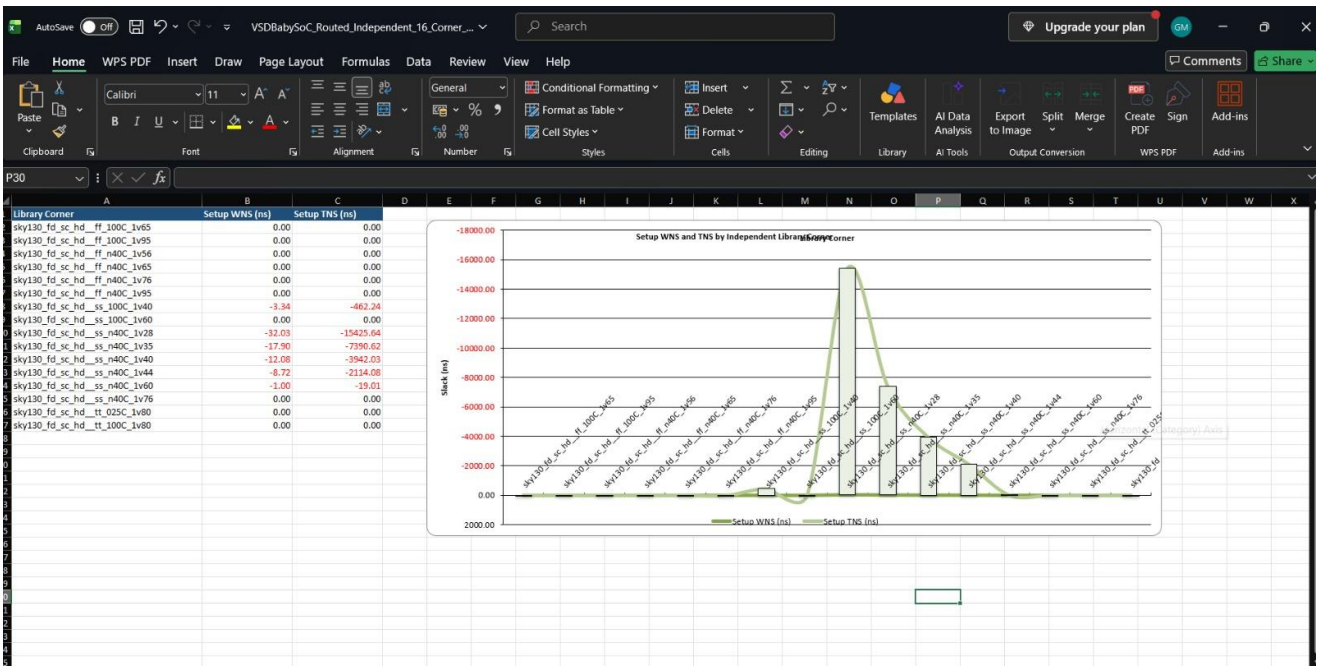
7. Graphs and presentation recommendations

The supplied combined setup chart is difficult to read because TNS can be thousands of nanoseconds negative while WNS is only tens of nanoseconds negative. Plot the metrics separately.

Recommended charts:

- Setup WNS by corner — column chart, y-axis in ns.
- Setup TNS by corner — separate column chart, y-axis in ns.
- Worst setup-path slack by corner — separate column chart.
- Worst hold-path slack by corner — separate column chart.
- If plotting hold WNS/TNS, show them in a separate chart and explain that all-zero values mean no negative hold slack wasreported, not that worst hold-path slack equals zero.

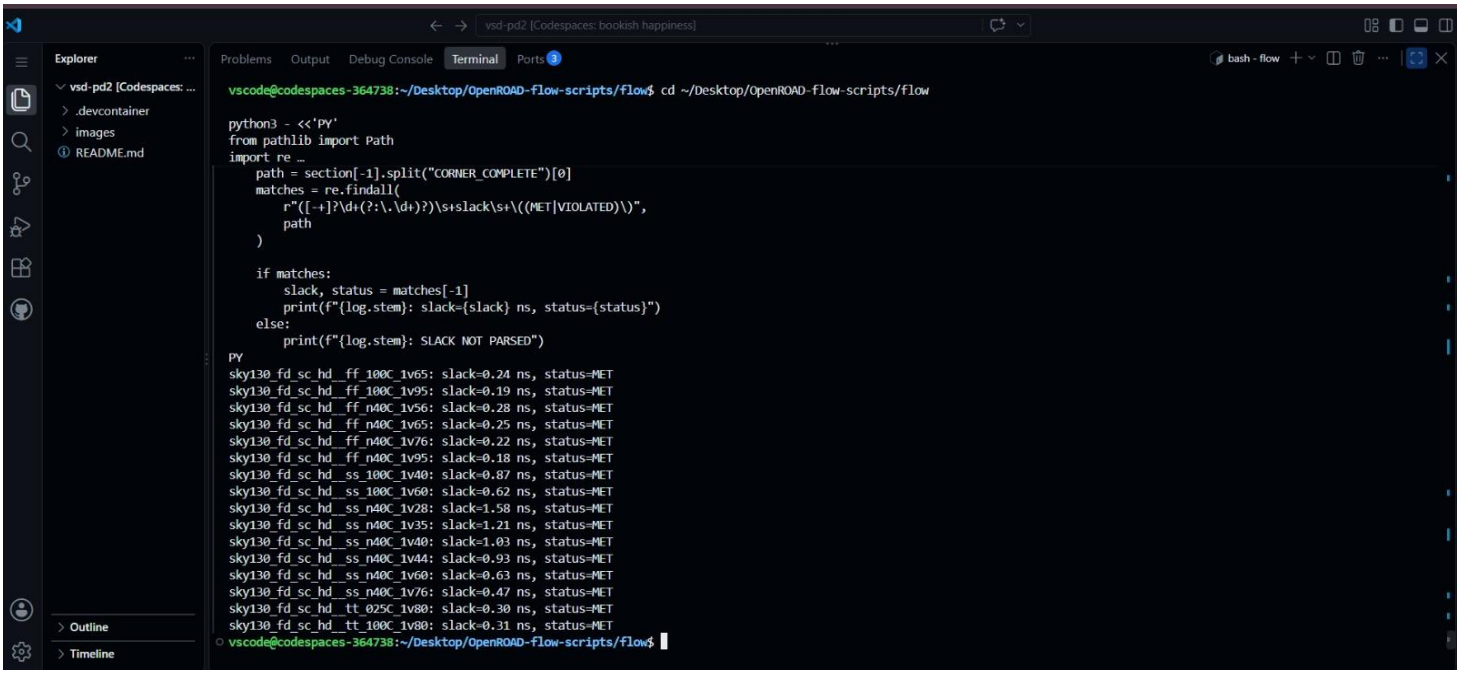
Use the full library-corner name as each category label. For readability, rotate category labels about 45° or use shorter labels such as `ss_n40C_1v28`, while retaining a full-name lookup table.

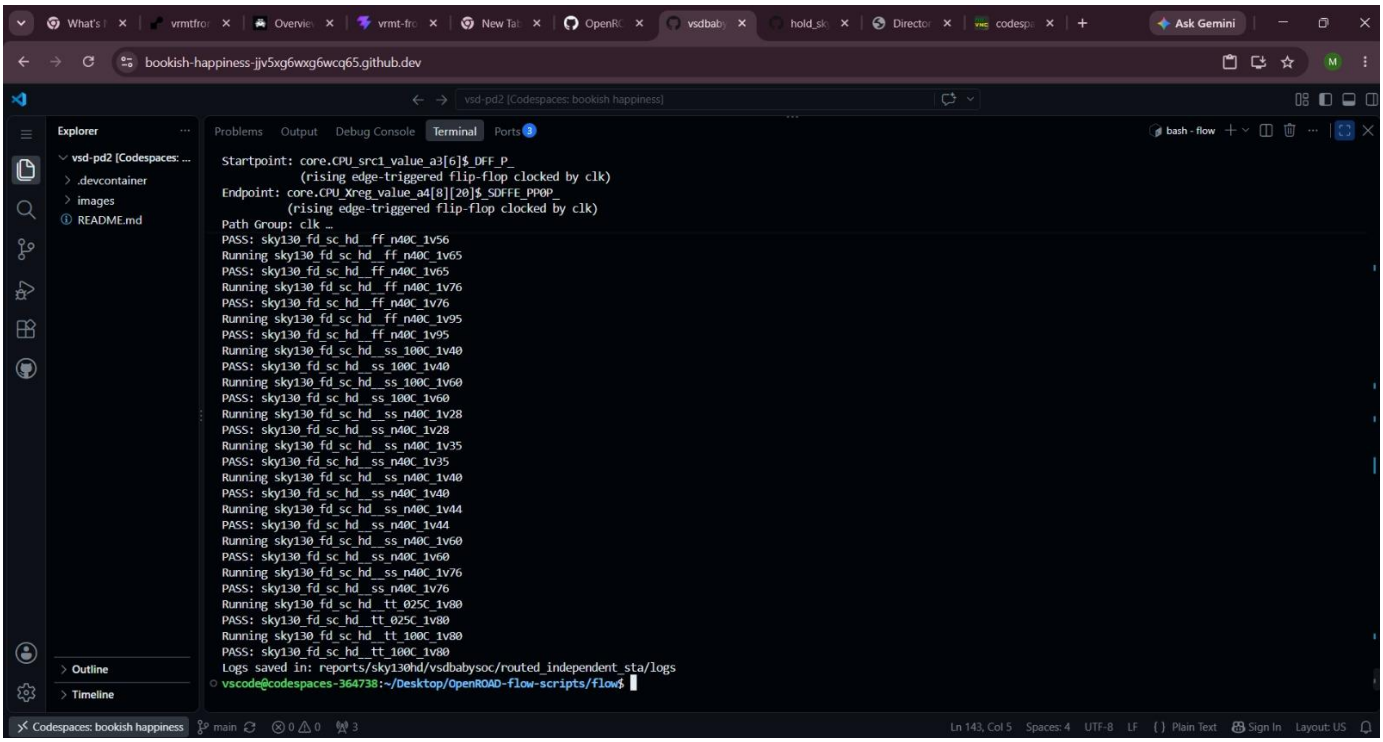


Library Corner	Setup WNS (ns)	Setup TNS (ns)
sky130_fd_sc_hd__ff_100C_1v65	0.00	0.00
sky130_fd_sc_hd__ff_100C_1v95	0.00	0.00
sky130_fd_sc_hd__ff_n40C_1v56	0.00	0.00
sky130_fd_sc_hd__ff_n40C_1v65	0.00	0.00
sky130_fd_sc_hd__ff_n40C_1v76	0.00	0.00
sky130_fd_sc_hd__ff_n40C_1v95	0.00	0.00
sky130_fd_sc_hd__ss_100C_1v40	-3.34	-462.24
sky130_fd_sc_hd__ss_100C_1v60	0.00	0.00
sky130_fd_sc_hd__ss_n40C_1v28	-32.03	-15425.64
sky130_fd_sc_hd__ss_n40C_1v35	-17.90	-7390.62
sky130_fd_sc_hd__ss_n40C_1v40	-12.08	-3942.03
sky130_fd_sc_hd__ss_n40C_1v44	-8.72	-2114.08
sky130_fd_sc_hd__ss_n40C_1v60	-1.00	-19.01
sky130_fd_sc_hd__ss_n40C_1v76	0.00	0.00
sky130_fd_sc_hd__tt_025C_1v80	0.00	0.00
sky130_fd_sc_hd__tt_100C_1v80	0.00	0.00

8. Validation and conclusion

Before treating these values as final signoff results, verify that each log records the expected Liberty file, routed ODB, extracted SPEF, and route SDC, and that the reported summary metrics were extracted from the intended timingcommands.





Important distinctions:

- A completed script marker confirms script completion, not necessarily correctness of every parsed value.
- WNS/TNS are summary metrics; worst setup/hold path slack is a separate metric.
- Independent one-library runs are useful for comparison, but the result is only meaningful when constraints, parasitics,clocks, and corner selection are consistent with the intended analysis.
- The reported setup failures at several slow-slow corners should be investigated before claiming timing closure.

Conclusion: the supplied data show setup violations in multiple slow-slow corners, with the most severe reported at `ss_n40C_1v28`. The supplied worst hold-path slack is positive and marked MET for every corner. The screenshots document floorplanning, placement, routing, and post-route timing analysis.